

Design and Implementation of a Role-Based Access Control (RBAC) Real-Time Location Management System using Firebase and Google Map APIs

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Abstract

Location-based services have now become an essential component, effectively visualizing geographic information to users through the web and modern mobile devices. However, existing platforms like Google Maps prioritize and focus on general purposes and navigation usage. Existing platforms operate in a restrictive manner and support limited role-based access, customized systems, and categorized geographic location handling. The aim of this paper is to develop a role-based geographic location management system to improve controlled data access and management for web and mobile platforms. The main objective of the project is to design and implement a system that allows administration to add, edit, and delete location data while restricted users are able to view categorized locations. Email notifications are used for account registration and password recovery to improve the user experience. Furthermore, the proposed system allows administrators to manually customize location markers using image URLs. The system is developed using Firebase for user authentication and real-time location management. Furthermore, Google APIs are integrated to visualize locations using user interfaces, which enhances the user experience by providing interactive maps and real-time updates on location data. To ensure usability and accessibility, the system is developed across web and mobile platforms using Android Studio and VS Code. The system was evaluated using criteria such as system performance, accuracy, and responsiveness. The findings revealed that the system updates made by administrators are instantly reflected among all restricted users, and role-based access control ensures secure location visibility. The outcome demonstrates that the proposed system not only improves role-based access control but also offers improvements over traditional location-based applications in terms of flexibility, responsiveness, and administrative control.

Keywords: *Firebase, Google Maps API, Location-Based Services, Real-Time Location Management, Role-Based Access Control*

1. Introduction

Location-based services are now used by people to interact with digital systems. People have become accustomed to accessing geographic information using tools like Google Maps. For instance, using location-based services to find a hospital in an emergency. These platforms are still built for general use and work well for navigation but are not good for situations where administrators need to control or manage data in a specific system. For instance, in a business or university setting, administrators might have to choose who can add locations and who can only see them. Therefore, the research problem lies in the lack of access-controlled geographic data management and data visualization systems. Many existing systems do not allow administrators to store geographical locations and do not allow restricted users to view those locations. The aim of this paper is to create a web and mobile geolocation management system with RBAC using Firebase and Google Maps APIs. The purpose of the system is to allow administrators to add, edit, delete locations, manage users, and allow a limited group of users to view locations. Features such as email notifications and flexible map marking are provided to enhance user experience.

2. Literature Review

Recent location-based services focus on secure user permission management and cross-platform access. For instance, Al Lail et al. (2024) described a way to secure data implementing the Generalized Spatio-Temporal RBAC Model (GSTRBAC). They proposed to provide access-based control on user credentials. As Eryonucu & Papadimitratos (2022) paper, many popular location-based services do not allow strong security and also make attacks easier. The developed system aims to bridge this gap by allowing administrators to manage geographic locations and allowing users with access to view locations. Recent studies validate Google Maps APIs and Firebase for real-time and cross-platform synchronization. M.S. Saran et al (2025) demonstrates that the road data collection (GetMap) Android and

Firestore supports users to upload and download collected field data simultaneously. They demonstrate that the Firestore is a cloud storage that includes Authentication, Real-time Database, Cloud Storage, and Cloud Firestore. Firestore's SDK is cross-platform, allowing projects to be integrated with Android, iOS, and web platforms. In practice, developers utilize Google Maps APIs and Firestore to effectively visualize live maps and map markers. For instance, Zakiyah Rahmanti et al (2022) developed a transportation system utilizing Google Maps APIs and Firestore as backend. Amik Lamappapoleonro Soppeng et al (2021) developed a location based public transportation web application using JavaScript. They demonstrate that JavaScript is able to work in most popular web browsers such as Internet Explorer. These works are supported to implement a real-time RBAC location management system with categorized location handling and instant synchronization utilizing Firestore and Google Maps.

3. Research Materials and Methods

This paper follows a system development approach to create a RBAC real-time location management system to solve a real-world problem. This system is developed using Firestore as backend for user authentication, real-time synchronization, and data storage. Firestore store location data, user data, and marker URLs in JSON format. Google Maps APIs such as Maps SDK for Android and Maps JavaScript API for web are integrated to provide interactive interfaces for visualizing map markers. The system is developed using Bootstrap to be compatible with both web and mobile devices to ensure accessibility across devices. RBAC has been implemented to allow administrators to manage geographic data and allow users to view the data via web or mobile applications. Features such as email notifications, responsive web design, and password recovery have been included to enhance usability. PHP Mailer was used to send email notifications. Web pages and mobile applications have been implemented using Android Studio and VS Code IDEs. Functional testing is conducted to evaluate data accuracy, responsiveness, and system performance under different user scenarios.

Table 1: Research Materials

Research Material	Purpose
Firestore	Authentication and Real-time Database
Google Map APIs	Visualizing map and map markers
PHP Mailer	Password recovery and registration notifications
Bootstrap	Responsive web design
VS Code	Website creation
Android Studio	Mobile application creation

4. Results and Discussion

The implemented system demonstrates that a RBAC real-time location management approach can be effectively achieved using the Google Maps API and Firestore. During the testing, users were able to log in to the system without any difficulty. Administrators could easily add, edit, delete locations, and manage users. Furthermore, restricted users were able to view those updates immediately. For example, when a new laboratory location was added, it could be viewed instantly within seconds using the website or mobile app. The real-time RBAC worked as expected. Allow administrators to insert, edit, and delete new data, as well as users to securely view data and prevent external parties from accessing that data. Email notifications such as registration and password recovery worked reliably, thereby improving the user experience. The system showed low response time during normal usage and delivered accurate results. However, performance with large-scale datasets was not thoroughly evaluated, indicating a need for further testing in future work.



Figure 1: Login Page



Figure 2: Home Page



Figure 3: Add Location Page



Figure 4: Add Member Page



Figure 5: Login page (mobile application)



Figure 6: Location Viewing page (mobile application)

5. Conclusion/s and suggestions

The developed system demonstrates that an efficient and effective geolocation management system can be implemented using a RBAC method with the help of Firebase and Google Maps API. Basic tasks such as handling geolocation data and managing users can be performed without difficulty, and updated data is reflected instantly with the help of web and mobile applications. This kind of responsiveness makes the system feel reliable from a user's point of view. Managing locations based on categories simplifies navigation, and its importance depends on how administrators use categories. This system ensures that users only have the ability to view data, while administrators have the ability to modify and view data. The user experience has been improved by implementing user registration email notifications and password recovery email notifications. As a future work, search and filtering options based on categories can be developed so that users can handle large amounts of data without any hassle. The system can be further enhanced by adding features such as location-based notifications and analytics. Overall, the system solves a real-world problem, but it is not the final step.

6. References

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